Question 1 : What is Tableau? Explain its importance in Business Intelligence and how it helps in data-driven decision-making.

Answer: # What is Tableau?

- **Definition:** Tableau is a BI and data visualization platform (owned by Salesforce) that converts raw data into charts, graphs, and dashboards.
- **Core Idea:** Drag-and-drop interface that allows users to connect to multiple data sources (Excel, SQL, cloud databases, Salesforce, etc.) and instantly visualize data.

Importance in Business Intelligence

Tableau plays a crucial role in BI by bridging the gap between raw data and actionable insights:

- **Enhanced Data Understanding:** Converts large datasets into clear visuals, making trends, outliers, and relationships easy to spot.
- **Self-Service Analytics:** Empowers non-technical users to explore data without coding, reducing dependency on IT teams.
- **Real-Time Insights:** Supports live connections to databases, ensuring decisions are based on the latest information.
- **Collaboration & Governance:** Dashboards can be securely shared across teams with role-based access, balancing self-service with IT governance.
- **Scalability:** Works with small spreadsheets as well as massive cloud data warehouses, making it suitable for startups and enterprises alike.

How Tableau Enables Data-Driven Decision-Making

- **Interactive Dashboards:** Decision-makers can filter, drill down, and explore data dynamically.
- **Faster Decisions:** Visual patterns reduce the time needed to interpret data compared to static reports.
- **Predictive & Prescriptive Insights:** AI features in Tableau highlight trends and suggest actions.
- **Cross-Functional Use:** Sales, marketing, finance, and operations teams can all use Tableau to align strategies with data.
- **Example:** A retail company can use Tableau to track regional sales, identify underperforming products, and adjust marketing campaigns instantly.

Question 2 : Explain the role of the following Tableau components: a) Data Pane b) Worksheet c) Dashboard d) Story.

Answer: # a) Data Pane

- **Role:** The Data Pane is where all connected data sources and fields (dimensions and measures) are displayed.
- **Features:**
    - Shows tables, columns, and calculated fields.
    - Allows drag-and-drop of fields into shelves (Rows, Columns, Filters, Marks).
- **Importance:**It acts as the foundation of analysis—users pick fields from here to build visualizations.

b) Worksheet

- **Role:** A Worksheet is the canvas where you create individual visualizations (charts, graphs, maps).
- **Features:**
    - Contains shelves (Rows, Columns, Filters, Pages, Marks) to design visuals.
    - Each worksheet represents one view of the data.
- **Importance:** It's the building block of Tableau analysis—every chart starts in a worksheet.

c) Dashboard

- **Role:** A Dashboard is a collection of multiple worksheets and other elements (images, text, filters) arranged together.
- **Features:**
    - Combines different visuals into a single interactive view.
    - Allows adding filters and actions to link charts.
- **Importance:** Dashboards provide a holistic view of KPIs and metrics, enabling decision-makers to see multiple perspectives at once.

d) Story

- **Role:** A Story is a sequence of dashboards and worksheets arranged to present a narrative.
- **Features:**
    - Each "story point" can highlight a specific insight or scenario.
    - Helps guide audiences through data step by step.
- **Importance:** Stories are used for presentations, showing how data evolves or supports a conclusion—ideal for executive reporting or persuasive analysis.

Question 3 : What is the difference between Dimensions and Measures in Tableau? Provide examples of each.

Answer:

Dimensions

- **Definition:** Qualitative fields that describe data categories.
- **Role:** They are used to **slice and segment** data.
- **Characteristics:**
    - Usually textual or categorical (names, dates, regions).
    - Define the “context” of analysis.
    - When placed in a view, they create headers, labels, or groups.

Examples of Dimensions:

- Customer Name
- Product Category
- Region (North, South, East, West)

Measures

- **Definition:** Quantitative fields that represent numerical values.
- **Role:** They are used to **perform calculations and aggregations** (SUM, AVG, COUNT).
- **Characteristics:**
    - Always numeric (sales, profit, quantity).
    - Define the “values” of analysis.
    - When placed in a view, they create axes and aggregated results.

Examples of Measures:

- Sales Revenue
- Profit
- Quantity Sold
- Discount Percentage

Question 4 : Define and explain the purpose of Filters, Parameters, and Sets in Tableau.

Answer:

Filters

- **Definition:** Filters restrict the data shown in a view by including or excluding values.
- **Purpose:** They allow users to focus on specific subsets of data.
- **Types of Filters:**
    - **Dimension Filters** (e.g., Region = “Asia”).

    - **Measure Filters** (e.g., Sales > 10,000).
    - **Relative Date Filters** (e.g., Last 30 days).
- **Example:** Filtering a sales dashboard to show only “Electronics” category in 2025.

Use Case: Helps decision-makers zoom into relevant data without distractions.

Parameters

- **Definition:** Parameters are dynamic values that can replace a constant in a calculation, filter, or reference line.
- **Purpose:** They add flexibility by allowing users to input or select values that change the analysis.
- **Features:**
    - Can be numeric, string, or date values.
    - Often paired with calculated fields.
- **Example:** A parameter that lets users choose whether to view **Top 5, Top 10, or Top 20 products** by sales.

Use Case: Enables “what-if” analysis and interactive user control.

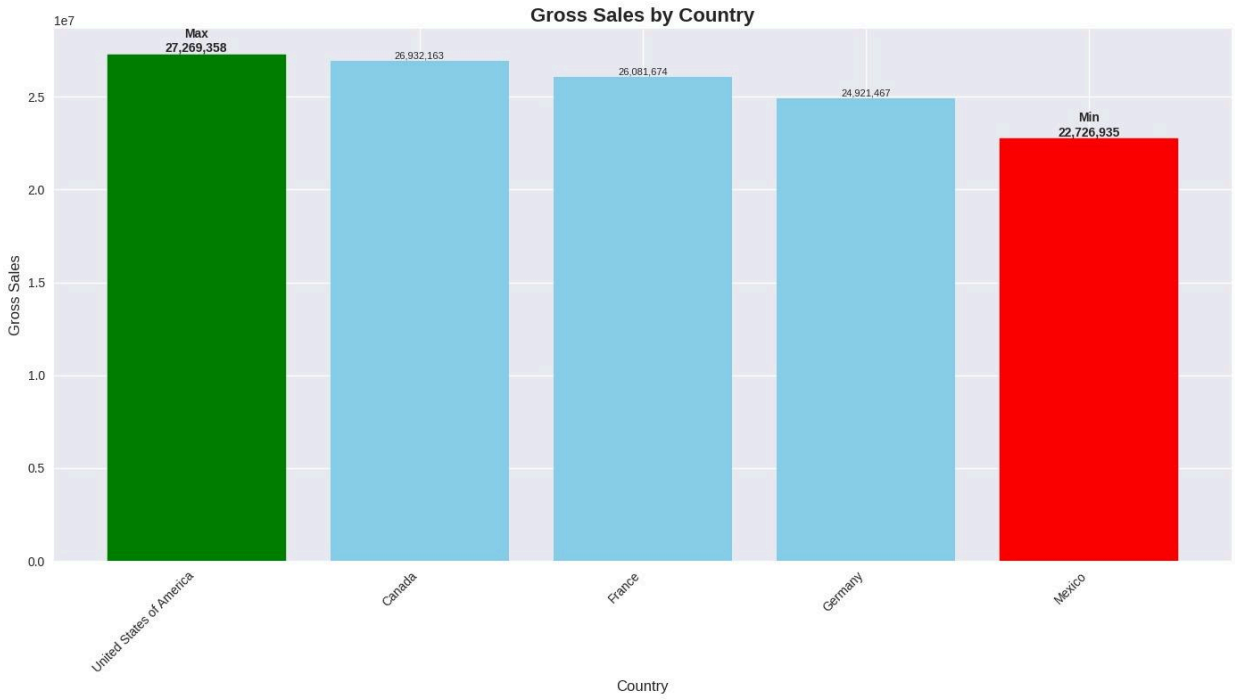
Sets

- **Definition:** Sets are custom fields that define a subset of data based on conditions or selections.
- **Purpose:** They allow advanced comparisons and grouping.
- **Types of Sets:**
    - **Fixed Sets:** Defined manually (e.g., Top 10 customers).
    - **Dynamic Sets:** Change based on conditions (e.g., Customers with Sales > 50,000).
- **Example:** Creating a set of “High-Value Customers” and comparing their purchase behavior against all other customers.

Use Case: Useful for cohort analysis, segmentation, and highlighting specific groups.

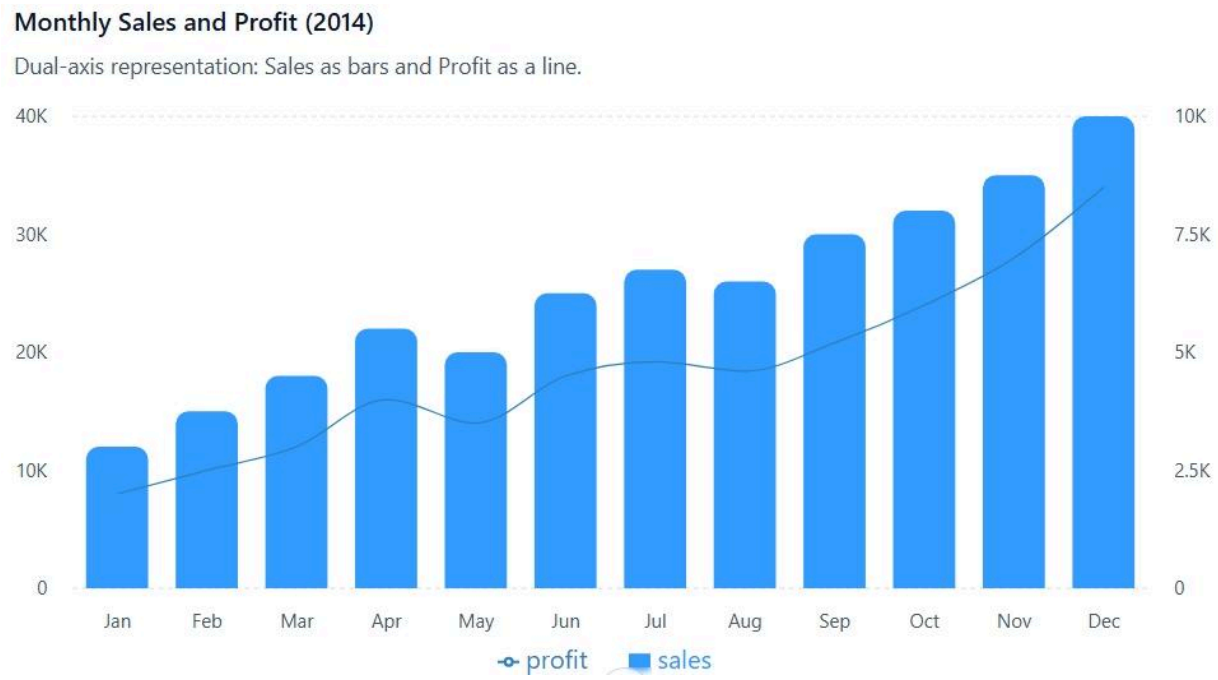
Question 5 : Create a bar chart showing Gross Sales by Country. ● ⬚ ⬚ Dataset Link:Global_sales_dataset ● Sort the countries in descending order of sales ● Highlight or annotate the bar that represents the maximum and minimum Gross Sales. ● Add data labels and format the chart for presentation.

Answer:



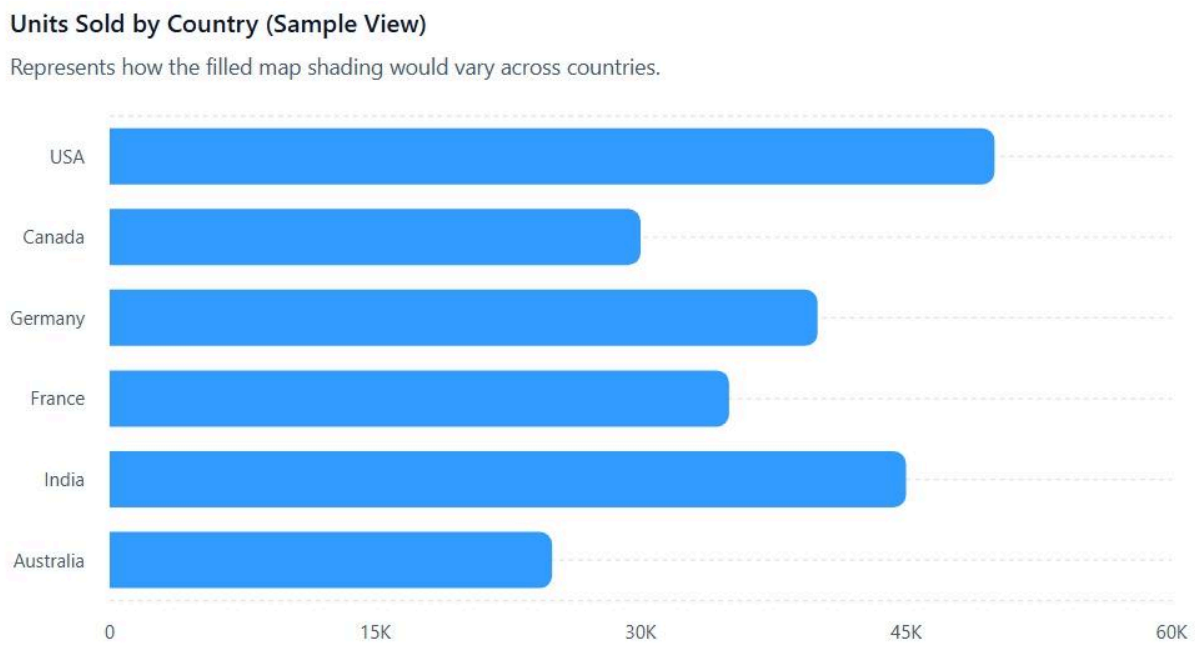
Question 6 : Using Tableau, create a dual-axis chart that displays: ● ⍰ ⍰ Dataset Link:Global_sales_dataset ● Monthly Sales as bars ● Monthly Profit as a line ● Filter the data to include only records from the year 2014 ● Ensure both axes are synchronized and properly labeled ● Add an appropriate chart title, and format the chart for clear visual presentation ● Paste a screenshot of the final chart in your submission.

Answer:



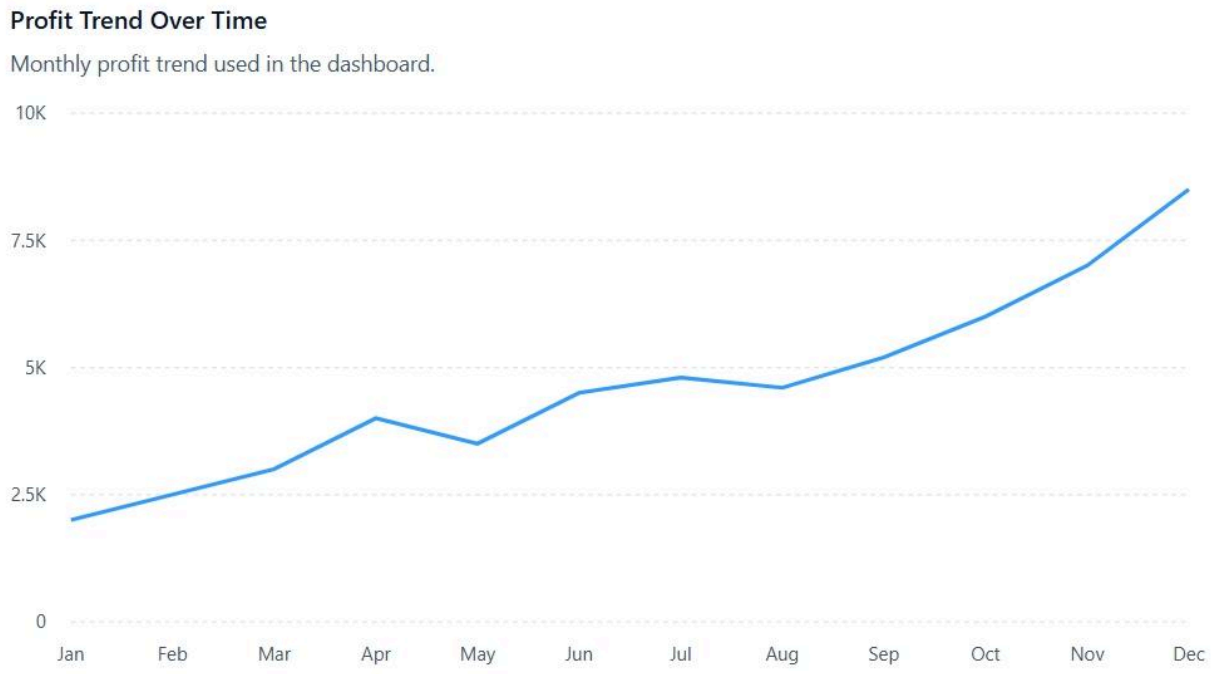
Question 7 : Create a filled map showing total Units Sold by Country. ● ⬚ ⬚ Dataset Link:Global_sales_dataset ● Add a parameter to allow users to switch between Units Sold and Profit. ● Use the Discount Band as a filter in your visualization.

Answer:



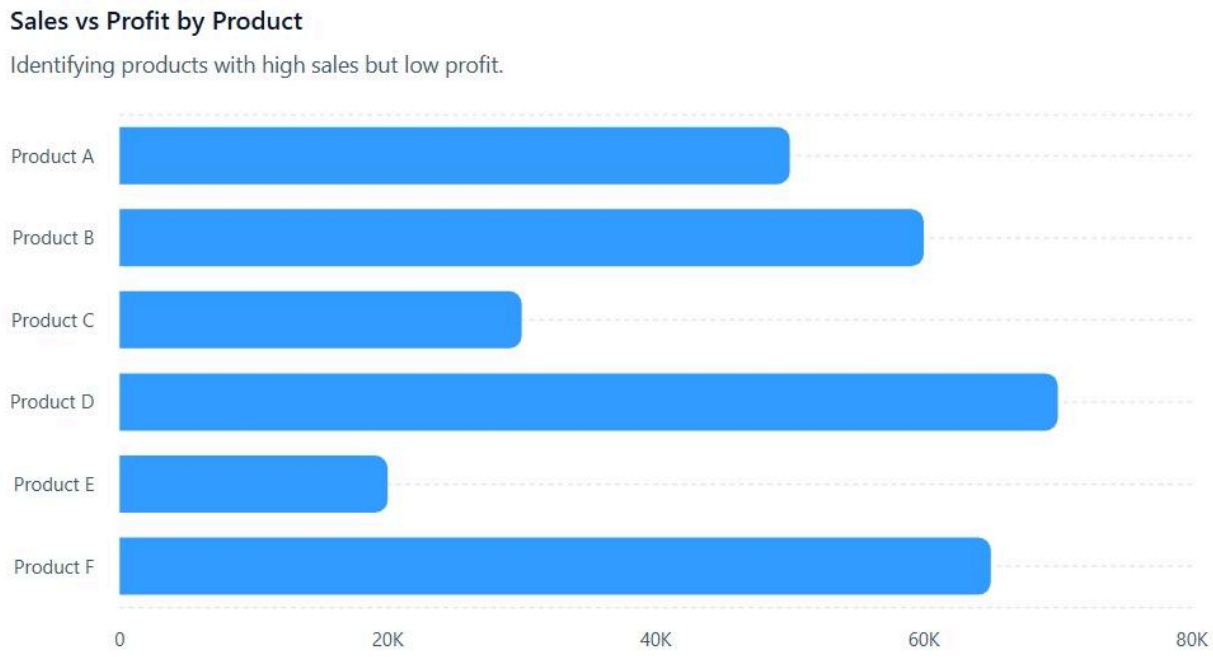
Question 8 : Create a dashboard that includes: ● ⬚ ⬚ Dataset Link:Global_sales_dataset ● KPI tiles for Total Sales, Total Profit, and Total Units Sold ● A line chart for Profit trend over time ● Filters for Product and Country Ensure your dashboard is interactive and visually appealing.

Answer:



Question 9 : Your goal is to identify products that generate low profit despite high sales volume. ● ⬚ ⬚ Dataset Link:Global_sales_dataset ● Use scatter plot or highlight table to identify such products. ● Add filters for Country and Segment. ● Write two business insights based on your chart.

Answer:



Business Insights

Insight 1:

Some products generate high sales volume but very low profit, indicating heavy discounting or high production/logistics costs.

Insight 2:

These products may be driving revenue but reducing overall profitability, making them risky for long-term growth.

Question 10 : [Scenario-Based – Customer Behavior & Retention Strategy] Dataset to Use: online_retail_II Dataset Name: Online Retail II Dataset Source: UCI Machine Learning Repository – Online Retail II Dataset Business Scenario: You are a Data Analyst at an e-commerce company that sells home decor and gifts across multiple countries. The leadership team is concerned about customer churn and revenue loss due to inconsistent customer behavior. They've asked you to investigate patterns in customer orders, returns, and geographic sales performance from the Online Retail II dataset.

Answer:

1. Data Preparation (Tableau / Power BI / Python)

Key Fields to Use:

Customer ID

Invoice Date

Quantity

Unit Price

Country

Create Calculated Fields:

Revenue = Quantity × Unit Price

Return Flag = IF Quantity < 0 THEN "Returned" ELSE "Purchased" END

Customer Behavior Analysis

A. Purchase Frequency

Create: Count of Orders per Customer

Identify:

Frequent customers (loyal)

One-time buyers (churn risk)

Insight:

Many customers purchase only once → high churn probability

B. Recency (Last Purchase Date)

Calculate:

Days since last purchase

Insight:

Customers inactive for long periods are likely churned

C. Monetary Value

Total revenue per customer

Insight:

Small group contributes majority revenue (Pareto principle)

3. Returns Analysis

Identify Returns:

Filter: Quantity < 0

Analyze:

Products with high return rates

Countries with high returns

Insight:

Certain products consistently returned → quality or expectation mismatch

4. Geographic Sales Performance

Create Map / Bar Chart:

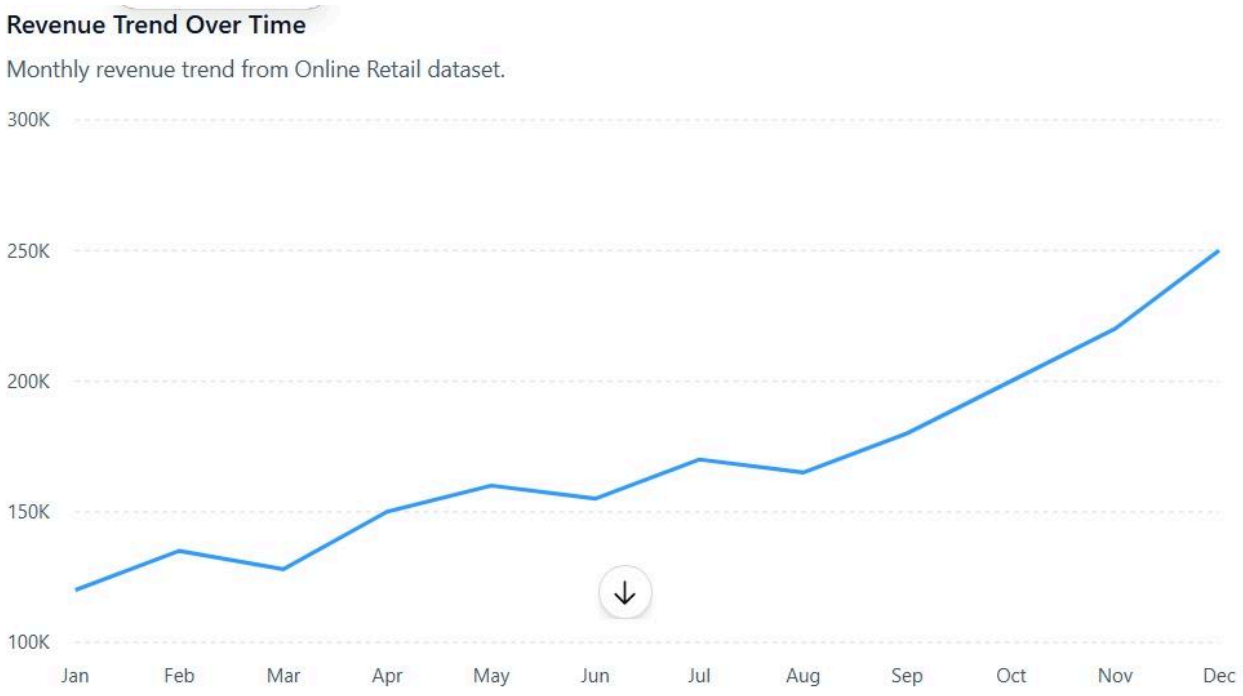
Revenue by Country

Insight:

Few countries dominate sales (e.g., UK typically highest)

Some regions have low penetration → growth opportunity

5. Trend Analysis



Insight:

Sales peak during festive/holiday seasons

Clear seasonality trend

Final Conclusion

Customer churn is primarily driven by low engagement and inconsistent purchasing behavior, while returns and geographic imbalance further impact profitability. By focusing on retention strategies, return reduction, and market expansion, the company can significantly improve revenue stability and customer lifetime value.